GRIFFIN LOWELL

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EDUCATION

University of Wisconsin–Madison — Madison, WI **Expected May 2028**

Bachelor of Science, Mechanical Engineering | Certificate in Computer Sciences
GPA: 3.41/4.00 | Dean's Honor List (2 terms)

Coursework: Dynamic Systems, Fluid Mechanics, Computer-Aided Engineering, Mechanics of Materials, Design of Machine Elements, Linear Algebra and Differential Equations, Intro to Artificial Intelligence, Human-Computer Interaction, Programming II and III (Java), Data Science Programming (Python)

SKILLS

Software: SolidWorks (CSWA Certified), FEA, MATLAB, Python, Java, Git, Excel
Shop and Hardware: Carbon DLS, HP Multi Jet Fusion, FDM 3D printing, laser cutting, mill, lathe, CNC lathe, bead blasting, vapor smoothing, Raspberry Pi, soldering

EXPERIENCE

Engineering Intern — The Technology House, Streetsboro, OH **May 2026 – Aug 2026**

- Designed and built a mobile powered fixture that lifts, tilts, and rotates a Carbon resin cassette outside the printer, eliminating the leak path that drained resin into machine internals and replacing an 18 min scooping routine with a clean corner pour
- Delivered the fixture and presented the design, test results, and a proposed containment fix to the engineering team
- Operated Carbon DLS printers daily, measuring resin pours to spec, cleaning machines, washing and baking finished parts, and performing quality control on completed builds
- Ran HP Multi Jet Fusion production end to end, including printer cleaning and maintenance, part extraction from cooling units, build unit material loading, bead blasting, and vapor smoothing

Head of Sailing — Camp Kawaga, Minocqua, WI **June 2023 – Aug 2025**

- Oversaw daily sailing instruction and free sail for campers, maintaining on-water safety and coordinating an inter-camp regatta

PROJECTS

Spread Trader Bot — Python, Charles Schwab API **May 2026 – Present**

- Built an autonomous options-spread trading system that scans live option chains, prices paired credit spreads, and routes real orders with real capital through the Charles Schwab API
- Raised daily fill volume from roughly 30 trades to over 140 and peak daily collected credit to $1,460 by retuning midpoint pricing and a sell-to-rescue retry loop after live fills lagged dry-run projections
- Tracks every filled trade and its collected credit through an automated logging pipeline; the bot runs live and continues trading daily
- Diagnosed an authentication failure that silently killed the live bot 25 minutes into every run, isolating overlapping trading sessions as the cause after ruling out a suspected library bug

Rotary Inverted Pendulum — SolidWorks, Raspberry Pi, Python **May 2026 – Present**

- Building an underactuated 2-DOF balancing system to implement energy-based swing-up and LQR stabilization, holding a passive pendulum inverted through a single motor driving a horizontal arm
- Modeled and 3D printed the base, driven arm, pendulum, and sensor housings; integrated a Raspberry Pi Zero 2 W, L298N H-bridge, N20 quadrature-encoder gearmotor, and AS5600 12-bit magnetic encoder over I2C
- Specified a software PWM duty ceiling to run the 6 V motor safely off a 12 V rail; mechanical and electrical build complete, controller implementation next

Raspberry Pi Door Buzzer — SolidWorks, 3D printing, Python **Oct 2025 – Dec 2025**

- Designed a 3D-printed rack-and-pinion actuator for an apartment entry buzzer with Wi-Fi remote unlock through a Raspberry Pi server

LEADERSHIP AND ACTIVITIES

Zeta Beta Tau, Alpha Kappa Chapter — Rush Committee, Academic Chair **Sept 2024 – Present**

- Planned recruitment events for 150+ underclassmen and ran weekly study tables giving freshmen academic guidance

Campus Ambassador — Seat Stock, student ticket marketplace **Aug 2025 – Present**

Interests: Marathon running (3:28 Madison Marathon), weightlifting, nutrition, DJ-ing